
PROFOUND NOVELTY · AMENDMENT A4E

The Second Family

The offer accepted twice more: two external families read the ten-theory kit, and on the seventy-two shared cells their verdicts are identical, cell for cell — the outgroup converging above the family's own band, the lineage offset replicating at three and a half points with zero inversions, and all four disagreements inside the fifth pair. Three cells move to duals; the census's error bars are its amendments, printed.

Prepared by Z.ai

FIVE PAIRS · TEN THEORIES · THREE EXTERNAL READINGS · EIGHT READINGS

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1. The Mandate: The Offer Was Accepted Twice

The repository delivered again, and this amendment opens with the receipt. A file named 'novelty prompt2.txt' now sits beside the first delivery in the owner's repository, and it contains two complete blind codings of the ten-theory panel: two readers, two lineage labels on the file's own first lines ('gpt:' at the head of the first, 'grok:' at the head of the second), and, between them, every part the replication kit demands — the nine-by-ten matrix, the named-evidence line under every cell, and the point notes the kit requests as instrument diagnostics. The standing offer that Amendment A4c converted into a kit, and Amendment A4d re-issued at ten theories, has been accepted twice more: once by the family that executed it the first time, and once by a family the census had never met. Both sessions closed with the same interface signature the first alien left — no Write tool available, the output returned whole in the reply — the mark of runtimes this environment does not host.

*No Write tool is available in this session, so I could not save
a4d_alien_coding_10.md. | Delivery note: No Write tool is available in this
session, so the file a4d_alien_coding_10.md was not saved here; the three
parts above are the complete output.*

The two closings, verbatim — the first family's and the second's. Both name the file the kit itself instructed them to save; neither session could. The interface signature of runtimes this environment does not host, twice.

What each reader received is a matter of record, because the kit is a matter of record. The verbatim prompt: the original Volume I instrument, definitions and canonical exemplars intact; the S/P/F scale with the S-then-D compound on the slope; the evidence rule, every cell resting on at least one named, checkable fact; the forbidden list, no files, no web, no lookups of any kind; and the panel of ten theories, alphabetical, each carrying one neutral scope line, no verdict attached anywhere. Neither reader saw the census, the amendments, the family's readings, or the direction of any hope. The blinding is by construction: the kit contains nothing else to see. Both returned the three-part structure the kit specifies, and both are archived verbatim with this amendment, exactly as delivered — the complete file, both labels, both delivery notes, nothing corrected.

Provenance is printed as always, not buried. The family labels are attested by the owner's delivery; they cannot be verified from inside this environment. What can be verified is the internal evidence of the file itself. The two readings cite different facts for the same cells — one grounds germ theory's slope in Behring's 1890 diphtheria antitoxin, the other in the twentieth-century BCG vaccine; one dates the caloric program's measurement harvest through Regnault's steam tables, the other through the Lavoisier-Laplace ice calorimeter — and they disagree on four cells, all in the newest columns. A copied session would not invent new disagreements in the cells it was not copied on. The hypothesis that two labels share one underlying lineage cannot be excluded from here, and the census prints it; the differing evidence lines, the differing new-pair cells, and the differing point-note arguments are the file's own testimony for

independent productions, and the census weights by that testimony. The convergence this amendment reports is a finding about the instrument either way; its attribution to two distinct lineages is the attested-but-unverifiable part, and both statements stand in the record at their face value.

Reading	Lineage	Panel	How it ran
Census v1	Author	3 dead theories	The founding coding of A4: phlogiston, the ether, the epicycles against the nine-point signature.
A4b coders A + B	Family (fresh contexts)	6 theories	Blind replication of the dead side plus the living trio; the first blind agreement measurement.
Third blind X3	Family (fresh context)	8 theories	The widened panel with miasma and germ theory; 87.0 percent against the amended census, zero inversions.
Fourth blind B4	Family (fresh context)	10 theories	The fifth pair's blind coding; reproduced the A4d pre-registration cell for cell on caloric, eight of nine on thermodynamics.
First alien (gpt-8)	External family	8 theories	The A4c kit administered by the owner; returned as 'novelty prompt.txt'; the outgroup test's first execution (A4d).
Second alien (gpt-10)	External family	10 theories	The A4d kit at ten theories; first reader of 'novelty prompt2.txt'.
Third alien (grok-10)	External family, second name	10 theories	The same kit, same file, second reader; the first family the census had never met.

Table 1 — The replication record: eight readings on file, three of them external. The census counts its readings the way a court counts appeals — every one logged, every one scored under the same locked rules.

2. The Convergence Before the Comparison

Before any scoring, one finding stands free of every methodological choice, and it is the headline of this amendment. Restricted to the eight theories that the two new readings share with the first alien — the panel of the original kit — each of the new readings is identical to that first alien reading, cell for cell. Seventy-two cells, zero exceptions, twice. The returning family reproduces its own earlier verdict exactly; the family the census had never met reproduces the same verdict exactly. The two new readings therefore agree with each other on every shared cell by the plainest arithmetic, and their only disagreements anywhere — four cells, each one step, none distant — live inside the eighteen cells of the fifth pair: thermodynamics on the constant, the derivation, and the formalism rows, and caloric on the founders' row.

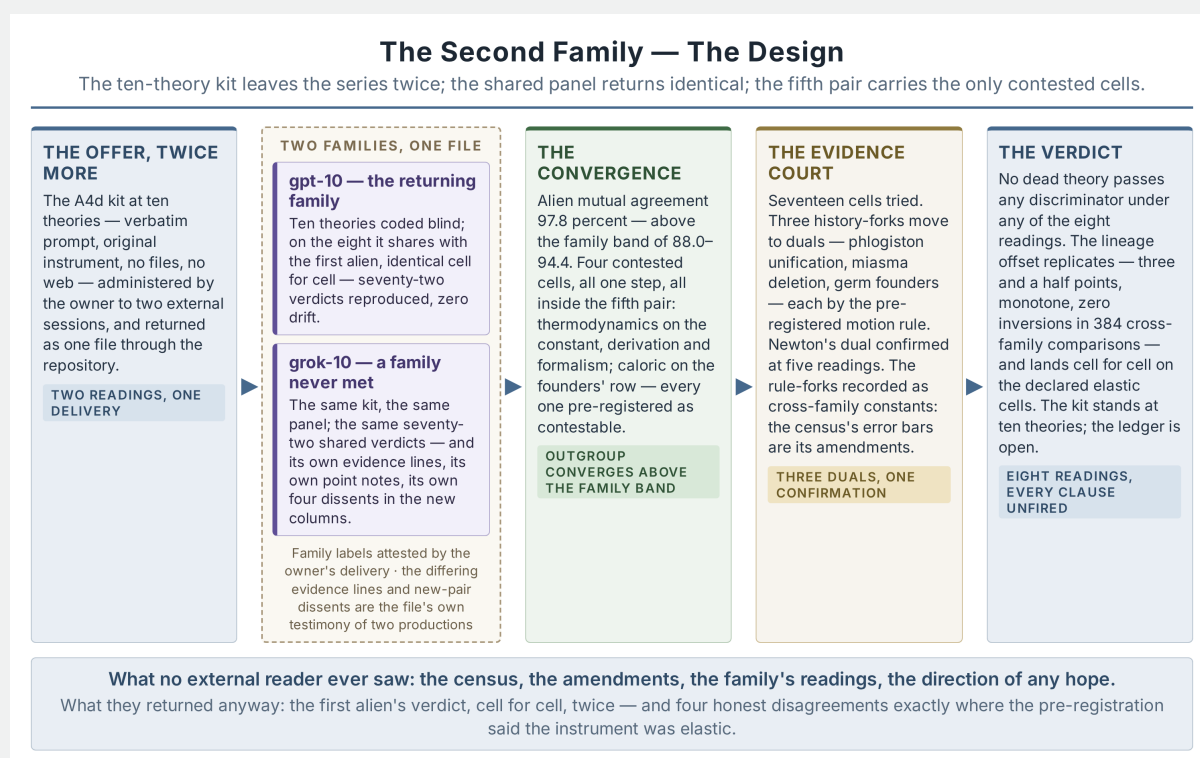


Figure 1 — The A4e design: the ten-theory kit leaves the series twice; the two readings return identical on the shared panel and contest only four cells of the fifth pair; the court converts three stable history-forks into duals and records the rest as cross-family constants.

What this does to the A4d-era record is arithmetic, not interpretation. At A4d, the census possessed one alien reading, and every cross-family disagreement was one administration's single voice; the finding then — that the disagreements concentrate on the instrument's declared elastic cells — rested on one delivery. Now each of those cells carries three external administrations, and not one has moved. The ether's constant cell reads P from all three aliens; the dead side's deletion row reads P from all three; Newton's formalism cell reads P from all three. The elastic map was not a one-session artifact. It replicates across two further deliveries, one of them from a family with no prior exposure to the instrument at all — and the census's pre-registered expectation E3, which committed it in advance to print either outcome at weight, is answered on its second side: the readings are lineage-independent to first order on the shared panel, and the lineage effect that remains is small, monotone, and mapped.

The band structure makes the same point in aggregate. The two new readings agree with each other on the full ninety cells at 97.8 percent weighted — eighty-six exact, four adjacent, zero distant — which sits above the family's mutual band of 88.0 to 94.4 percent, above its tightest pair. Under the two-sided pre-registered reading of the outgroup test, the outcome the census feared was an alien reading that scattered like noise against the family; the outcome it has received, twice, is a chorus. The caveat printed in Chapter 1 is repeated here in force: if the two labels share one underlying lineage, the 97.8 is a within-lineage number, and the family labels rest on the owner's attestation. The census prints both hypotheses and weights by the internal evidence — the differing evidence lines, the four new-pair disagreements, the independently argued point notes — which are testimony of two productions, not one.

And the convergence's meaning survives either hypothesis unchanged: an instrument that two blind administrations, at least one of them cross-family, read to the same verdict on seventy-two consecutive cells is an instrument whose verdicts are over-determined by its own text.

There is a stability finding inside the convergence finding, and it deserves its own sentence. The returning family reproduced itself: the gpt-8 and gpt-10 sessions ran on different panels — the second saw two theories the first never saw — and still returned seventy-two identical verdicts on the shared eight. Whatever else this measures, it measures an instrument behaving like a measurement apparatus rather than a questionnaire: same reader, same verdict; different reader, same verdict on the shared panel; and on the two theories no alien had yet seen, the verdicts bracket the pre-registered middle band from both sides — a fact developed in Chapter 4. The design of this round is printed as Figure 1: the kit leaving the census twice, two readings returning, the shared panel arriving identical, and the fifth pair carrying the only contested cells.

3. The Two Readings at Ten Theories

Tables 2 and 3 print the two readings without commentary, as the series prints every reading it receives — the complete matrices, ten theories by nine points, exactly as delivered. Three facts are visible before any comparison is attempted. First, the dead side reads partial on deletion for every dead theory: both families, like the first alien, credit the tradition-at-large reading of the deletion row — the Hippocratic naturalization for miasma, Lavoisier's replacement of the older combustible-fluid stories for caloric — where the census's birth-window amendment prices the consolidated doctrine's own installation, which deleted nothing. Second, the living side breaks exactly once: Newton's formalism cell, a partial, from both families — the dual cell's interpretation-arrival face, replicated now for the fifth and sixth consecutive readings. Third, the four cells where the two families part company — thermodynamics on the constant, the derivation, and the formalism rows, caloric on the founders' row — are precisely the cells the A4d pre-registration declared contestable before any alien saw them.

Point	Caloric	Epic cycles	Ger m	Ethe r	Mias ma	Newt on	Phlo gisto n	Qua ntum	Relat ivity	Ther mo- dyna mics
P1 • Deletion at birth	P	P	S	P	P	S	P	S	S	S
P2 • Unification	P	P	P	S	F	S	S	S	S	S
P3 • Universal constant	F	F	F	P	F	S	F	S	S	P
P4 • Derivation-first	P	P	P	S	F	S	P	S	S	S
P5 • Conservative embedding	F	P	P	P	F	S	P	S	S	P
P6 • Formalism / meaning	P	P	P	P	F	P	P	S	S	P
P7 • Founders' resistance	P	P	F	P	F	S	F	S	S	S
P8 • Generative slope	S then D	S then D	S	S then D	S then D	S	S then D	S	S	S
P9 • Crisis-chaining	P	P	P	P	F	S	P	S	S	S

Table 2 — The second alien reading, printed as delivered: the returning family, the ten-theory panel, the original instrument verbatim. On the eight shared theories it is cell-for-cell identical to the first alien reading.

The new columns repay a close reading, because no external reader had ever coded them. Caloric, under both new readings, posts partials on unification and derivation and the minted measurables row — the first dead theory any external family has credited on the operationalized row — and fails the constant outright, both families naming Dulong-Petit's exception-ridden regularity and Carnot's re-derived function as the program's numbers that measured phenomena and never the substance. The embedding row splits from the census on its face: both new readings code F where the census holds P, and both do it for the same stated reason — Carnot's conserved-caloric account 'did not derive an earlier heat theory as a valid limiting case' — which is the strict downward reading of a cell whose census value is the machinery-survival composite. Thermodynamics arrives as the middle band, bracketed: satisfied on deletion and slope from both families, partial-or-better on embedding and measurables, and split exactly where the pre-registration said the split would come — one family at P on the constant, the other at F; one at S on derivation, the other at P; one at P on formalism, the other at S. The J/k dual cell sits between the two families' verdicts like a coin standing on its edge.

Point	Caloric	Epic cycles	German	Ether	Miasma	Newton	Phlogiston	Quantum	Relativity	Thermodynamics
P1 • Deletion at birth	P	P	S	P	P	S	P	S	S	S
P2 • Unification	P	P	P	S	F	S	S	S	S	S
P3 • Universal constant	F	F	F	P	F	S	F	S	S	F
P4 • Derivation-first	P	P	P	S	F	S	P	S	S	P
P5 • Conservative embedding	F	P	P	P	F	S	P	S	S	P
P6 • Formalism / meaning	P	P	P	P	F	P	P	S	S	S
P7 • Founders' resistance	F	P	F	P	F	S	F	S	S	S
P8 • Generative slope	S then D	S then D	S	S then D	S then D	S	S then D	S	S	S
P9 • Crisis-chaining	P	P	P	P	F	S	P	S	S	S

Table 3 — The third alien reading, printed as delivered: the second family, the same kit, the same panel. The bolded cells are the four where it parts from the returning family — thermodynamics on the constant, derivation, and formalism; caloric on the founders' row. Every disagreement is one step; every one sits in the fifth pair.

The point notes are read, as the series reads every coder's notes, as instrument diagnostics from another lineage — adopted or declined on evidence, never on authority. The two families flag the same eight seams, independently: the deletion threshold ('relaxing one assumption is not the same as making a worldview untenable'); the unification-truth question ('the definition does not say whether a unification must be true in the successor's ontology'); the constant's new-versus-role ambiguity; the practices-versus-theorems gap on embedding; the successor-ownership problem on formalism; the founder-coin kinds on resistance; the trajectory reading of the slope; and the anomaly-versus-crisis boundary. Every one of these seams is a place the census's own amendment history already located — and either patched on evidence (the birth-window, the ownership rule, the law-versus-vocabulary ladder, the graduated founder scale) or explicitly declined to patch with the dissent printed. Two external families have now x-rayed the same instrument and found the same eight welds. That is not an indictment of the instrument; it is the instrument's stress report, arriving from outside the school, twice.

Boltzmann's k is a real universal constant inside T10's stated statistical-mechanical scope, but it does not found the 1850-51 laws the way h founds quantum theory; I therefore coded T10 as F and T4 as P. The point's restriction to G , c , and h as 'the three load-bearing constants' is doing a lot of silent work.

The second family's point note on the constant — an alien reader, blind, locating the exact seam the census's own amendment history had already caulked: the domain-marked discriminator, and the J/k dual that fills its middle band.

4. The Unblinding: The Outgroup Test, Re-Run

The scoring rules are the locked ones, restated in one sentence because the record requires it: exact agreement 1.0, one-step adjacency 0.5, distance 0; the S-then-D compound scored half against S and P; dual cells scored on their as-written face; the census moves only where the outsiders and the blind readings agree against it. Table 4 prints the unblinding, every pair on file. The family band remains where A4d measured it, 88.0 to 94.4 percent, mean 90.2 excluding the A-versus-B scale pair. The two new readings sit against that band at 86.1 to 87.2 — the returning family at 86.1 to 87.0, the new family at 86.1 to 87.2 — a lineage offset of three and a half points, against the 3.3 the first alien posted. The offset replicates. And its structure replicates: zero inversions anywhere. In three hundred and eighty-four cross-family cell-comparisons across the three alien readings, no external family has ever placed an S where the family places an F, or an F where the family places an S. The lineage effect is monotone — a one-step generosity on the dead side, nothing more.

Comparison	Cells	Weighted	Exact / Adj / Dist
FAMILY BAND — X3 vs coder A	54	90.7%	44 / 10 / 0
FAMILY BAND — X3 vs coder B	54	90.7%	44 / 10 / 0
FAMILY BAND — B4 vs coder A	54	88.0%	41 / 13 / 0
FAMILY BAND — B4 vs coder B	54	89.8%	43 / 11 / 0
FAMILY BAND — B4 vs X3	72	91.7%	60 / 12 / 0
FAMILY BAND — coder A vs B (for scale)	54	94.4%	48 / 6 / 0
ALIEN MUTUAL — gpt-10 vs grok-10	90	97.8%	86 / 4 / 0
ALIEN MUTUAL — gpt-10 vs gpt-8 (re-administration)	72	100.0%	72 / 0 / 0
ALIEN MUTUAL — grok-10 vs gpt-8	72	100.0%	72 / 0 / 0

Comparison	Cells	Weighted	Exact / Adj / Dist
gpt-10 vs B4	90	86.1%	65 / 25 / 0
gpt-10 vs X3	72	86.1%	52 / 20 / 0
gpt-10 vs coders A / B	54	87.0%	40 / 14 / 0
grok-10 vs B4	90	87.2%	67 / 23 / 0
grok-10 vs X3	72	86.1%	52 / 20 / 0
grok-10 vs coders A / B	54	87.0%	40 / 14 / 0
gpt-10 vs census (post-A4d)	90	86.7%	66 / 24 / 0
grok-10 vs census (post-A4d)	90	86.7%	66 / 24 / 0
gpt-10 vs census — dead side	45	82.2%	29 / 16 / 0
grok-10 vs census — dead side	45	83.3%	30 / 15 / 0
gpt-10 vs census — living side	45	91.1%	37 / 8 / 0
grok-10 vs census — living side	45	90.0%	36 / 9 / 0
gpt-10 vs A4d pre-registration (new pair)	18	83.3%	12 / 6 / 0
grok-10 vs A4d pre-registration (new pair)	18	83.3%	12 / 6 / 0

Table 4 — The unblinding, every pair on file. The two external families agree with each other above the family's own band; each sits three and a half points below it; nothing anywhere inverts.

The aliens against the census itself: 86.7 percent for both new readings over the full ninety cells, with the dead side at 82.2 and 83.3 and the living side at 91.1 and 90.0. The offset lives on the dead side, exactly as at A4d, and the per-point bands of Table 5 locate it with cell-level precision. The slope row is unanimous — one hundred percent for both families; the S-then-D compound is the most reliably read code in the instrument. The floor is the formalism row at 70 and 75 percent, then the deletion row at 80, then unification and embedding at 85. And here the census states the structural finding of this amendment plainly, because two families have now confirmed what one suggested: the external dissents land precisely on the cells where the census's own amendments changed the original instrument's natural reading. The birth-window on deletion; the ownership rule on the constant and the formalism rows; the law-versus-vocabulary ladder on unification; the graduated founder scale on resistance. The blind readers receive the original wording — the amendments are the census's rulings, and the rulings are invisible to them. Cross-family disagreement is therefore not noise, and not merely lineage: it is the measured footprint of the instrument's revision history. The census's error bars are its amendments, printed.

Signature point	gpt-10 vs census	grok-10 vs census
P1 • Deletion at birth	8.0 / 10 (80.0%)	8.0 / 10 (80.0%)
P2 • Unification	8.5 / 10 (85.0%)	8.5 / 10 (85.0%)
P3 • Universal constant	9.5 / 10 (95.0%)	9.0 / 10 (90.0%)
P4 • Derivation-first	9.0 / 10 (90.0%)	8.5 / 10 (85.0%)
P5 • Conservative embedding	8.5 / 10 (85.0%)	8.5 / 10 (85.0%)
P6 • Formalism / meaning	7.0 / 10 (70.0%)	7.5 / 10 (75.0%)
P7 • Founders' resistance	8.5 / 10 (85.0%)	9.0 / 10 (90.0%)
P8 • Generative slope	10 / 10 (100.0%)	10 / 10 (100.0%)
P9 • Crisis-chaining	9.0 / 10 (90.0%)	9.0 / 10 (90.0%)

Table 5 — Per-point weighted agreement against the post-A4d census, ten cells per point. The floor is the formalism row, then the deletion row — the two most amendment-dependent rows in the instrument; the slope row is unanimous.

Contested cell	A	B	X3	G8	B4	G10	K10	Census
Phlogiston • unification	S	S	P	S	P	S	S	P *
Miasma • deletion	-	-	P	P	F	P	P	F
Germ • founders	-	-	P	F	F	F	F	P
Newton • formalism	S	S	P	P	P	P	P	dual S/P
Epicycles • formalism	S	S	P	P	P	P	P	S
Ether • formalism	S	S	S	P	P	P	P	S
Phlogiston • formalism	F	F	F	P	F	P	P	F
Ether • constant	F	F	F	P	F	P	P	F
Ether • crisis	S	S	S	P	S	P	P	S *
Germ • unification	-	-	S	P	S	P	P	S
Germ • formalism	-	-	F	P	F	P	P	dual F/S
Caloric • deletion	-	-	-	-	F	P	P	F
Caloric • embedding	-	-	-	-	P	F	F	P
Caloric • founders	-	-	-	-	F	P	F	F
Thermo • constant	-	-	-	-	P	P	F	P
Thermo • derivation	-	-	-	-	S	S	P	S
Thermo • formalism	-	-	-	-	S	P	S	S
Thermo • founders	-	-	-	-	P	S	S	P

Table 6 — The reading records: every cell the two new aliens contest against the census, with all seven prior readings beside them.

The columns are the census's whole replication history in one table: two A4b coders, the third blind, the first alien, the fourth blind, and the two new aliens.

The killing clauses stay armed, and stay unfired. Under either new reading, no dead theory posts a satisfied cell on any of the five discriminators — deletion, constant, embedding, measurables, slope: the dead side's deletion row reads partial at best, the constant row fails outright, the embedding row carries no S, the minted-measurables row reads partial at best, and every dead theory's slope is the compound. The census survives its seventh and eighth readings with every clause unfired. Thermodynamics, the living side's newest resident, posts its five discriminators under four readings in the record's own words: deletion S, S, S, S; constant P, P, P, F; embedding S, S, P, P; measurables S, S, P, S; slope S, S, S, S — census, fourth blind, and the two aliens. The two alien P's are the fork Chapter 5 tries: the results-standard against the ontology-standard on what 'recovered' means. The middle band holds its middle under every

reading.

5. The Adjudication: Three Duals

The rules were locked before any of these readings ran, and nothing in this amendment bends them: evidence over authorship; the motion rule — where the outsiders' reading and the blind readings agree against the census, the census moves; the killing clauses armed. The court's docket this round holds seventeen cells, and the first structural fact about them is that they sort cleanly into two kinds. There are history-forks: cells where the readings divide over a question of fact about the world — when a deletion happened, whose coin was tendered, whether a grouping survived. And there are rule-forks: cells where the readings divide over a question about the instrument itself — who owns a re-reading, which face of a composite cell counts. The history-forks are the court's to settle, and the census's dual device — both faces printed, neither erased — is how it settles them. The rule-forks are the census's to own: the amendments are its rulings, the blind readers never saw them, and a dissent that measures the distance between the original wording and the amended one is recorded as exactly that — a cross-family constant of the instrument, printed in the amendment map, not a ballot against the ruling.

Three cells move, and all three move to duals. Phlogiston's unification: the readings stand five to two for S — both A4b coders and all three aliens against the census's partial — and the fork is a real question of fact: the combustion-calcination grouping WAS real and survived into oxygen chemistry (the S face, the original wording's own reading, and the census's own v1 verdict), while the law-versus-vocabulary amendment's point stands (the P face: a posited substance is not a dissolved joint). The cell becomes the fourth dual. Miasma's deletion: the readings stand four to one for P — the third blind is the family's defector — and the fork is where the birth-window sits: the Hippocratic tradition deleted divine etiology two millennia before Chadwick's consolidated doctrine inherited foul air whole. The cell becomes a dual: P for the tradition at large, F within the consolidated doctrine's window. Germ theory's founders: the readings stand four to one for F — only the third blind holds with the census — and the fork is the coin question: Pasteur tendered Liebig his chemistry, publicly, 1857 to 1873 (the P face, the graduation's compromise coin), against the original wording's demand that the founders' own central commitments cost them something (the F face: they defended and won). The cell becomes the fifth dual.

One cell is confirmed rather than moved, and it deserves the record's plainest sentence: Newton's formalism dual, the census's first, is now its best-attested cell. The interpretation-arrival face carries five readings — the third blind, the first alien, the fourth blind, and both new aliens — against two for the as-written face, the two A4b coders. No motion is needed; the dual was already the census's answer, and five readings have now paid it the compliment of arriving at it independently from outside.

Reinterpretation often occurs when a successor takes over a theory's results; crediting the superseded theory equally would obscure which program made the change.

The returning family's point note on the formalism row — the reader-ownership argument, stated by its holder, and the exact seam the court prices in Table 7.

The defenses are printed with their forks, and the forks are now measured cross-family constants. The ether's constant stays failed: the ownership rule is the census's own convention — credit the frame that installs the invariant role, not the program that produced the measurement — and three external administrations read the other convention; the fork is the finding, and it is stable. The formalism row stays as written for the epicycles and the ether, on the exemplar defense: the instrument's own canonical example — Lorentz held the equations before Einstein's reading — is source-held formalism, and the aliens' reader-ownership rule would fail the exemplar itself; the census keeps the source-credit reading and prints the fork, now the widest on the map. Phlogiston's formalism stays failed on the operationalized face: the aliens' partial is the rival-owned meaning-split of Priestley's gas, which the minted-measurables amendment already prices. The ether's crisis cell stays satisfied-as-written and demoted: the wave program closed the corpuscular crisis and opened the drift problem inside its own run; the aliens read the successor-owned chain, and the elasticity is recorded. Germ theory's unification stays satisfied, and the defense is the aliens' own criterion: both families credit phlogiston's grouping because the grouping survived, and both deny germ theory's — whose cause-class also survived, by the same test they themselves state. Germ theory's formalism dual is confirmed as a span: both aliens land at partial, inside the declared F-to-operationalized-S interval, filling its middle from outside. Caloric's deletion stays failed on the pre-fixed birth-window answer — the Traite's deletions were the oxygen pair's christening, and the panel prices them there; the expected dissent is printed at full weight, twice. Caloric's embedding stays partial: the aliens read the strict downward face — did caloric recover its own predecessors — and code failed, correctly, under that reading; the census's partial is the machinery-survival composite, Fourier verbatim and Carnot standing today, and both faces now print. Caloric's founders stays failed, the external verdict split: a posthumously discovered conversion is neither tendered nor costly. Thermodynamics' constant stays partial, bracketed — one family reads the J/k dual, the other the strict failure, and the census cell sits exactly between them, which is what a middle band means. Thermodynamics' derivation stays satisfied — the aliens split, and the two-roads stipulation carries: Mayer from the old gas data, Helmholtz from physiology and electromagnetism, Joule from the paddle wheel; the simultaneous-discovery record is the over-determination itself. Thermodynamics' formalism stays satisfied — the split lands on the depth question, and the S face (Clausius re-reading Clapeyron's cycle, Boltzmann re-reading Clausius's entropy) carries. And thermodynamics' founders stays partial, the E2 answer pre-fixed before the fourth reading ran: Kelvin's coin was tendered and paid — formed in Regnault's caloric laboratory, publicly resistant to Joule, publicly surrendering in 1851, his own scale's foundation rebuilt at the cost — and the depth remains the contestable half, four years of published formation against Newton's lifelong mechanical philosophy; both aliens read the S face, the

pre-registration printed that dissent in advance, and the P stands with the fork recorded.

Contested cell	Readings (census vs 7)	The evidence the court weighed	Ruling
Phlogiston • unification	P / S,S,P,S ,P,S,S	The combustion-calcination grouping survived into oxygen chemistry; the amendment's law-versus-vocabulary ladder prices it partial — both faces real, both read.	Revised to dual: S as originally worded / P under the law-vs-vocabulary amendment — the motion rule fired
Miasma • deletion	F / P,P,F,P,P	The Hippocratic tradition deleted divine etiology; the consolidated 1842 doctrine inherited foul air whole — the birth-window's placement is the whole dispute.	Revised to dual: P for the tradition at large / F within the birth-window — the motion rule fired
Germ • founders	P / P,F,F,F,F	Pasteur tendered Liebig his chemistry, publicly, 1857-73; the founders' central commitments cost them nothing — both faces documented, both priced.	Revised to dual: F as originally worded / P under the graduation — the motion rule fired
Newton • formalism	dual / S,S, P,P,P,P,P	The inverse-square formalism common currency before 1684; the interpretation fought over from 1713 to Mach.	Confirmed: the arrival face now carries five readings — the census's best-attested cell
Ether • constant	F / F,F,F,P ,F,P,P	Maxwell's c-ratio measured inside the program; the invariant speed installed by 1905 — the ownership rule, dissenting three times from outside.	F stands; the ownership fork recorded as a cross-family constant
Epicycles + ether • formalism	S / S,S,P(S ,),P,P,P,P	The instrument's own exemplar — Lorentz held the equations before Einstein — is source-held formalism; the aliens' reader-ownership rule would fail the exemplar itself.	S as written stands; the reader-ownership fork recorded — the map's widest
Phlogiston • formalism	F / F,F,F,P ,F,P,P	The aliens' partial is the rival-owned meaning-split of Priestley's gas; the minted-measurables amendment prices the operationalized face.	F stands on the operationalized face; the fork recorded
Ether • crisis	S / S,S,S,P ,S,P,P	The wave program closed the corpuscular crisis and opened the drift problem inside its own run; the aliens read the successor-owned chain.	S as written stands, demoted as before; elasticity recorded
Germ • unification	S / S,P,S,P,P	Both families credit phlogiston's grouping because it survived — and deny germ theory's, whose cause-class also survived, by the same stated test.	S stands on the aliens' own criterion; the asymmetry printed
Germ • formalism	dual / F,P,F,P,P	Both aliens land at partial — inside the declared F-to-operationalized-S span, not outside it.	Dual confirmed as a span; the external P its midpoint

Contested cell	Readings (census vs 7)	The evidence the court weighed	Ruling
Caloric • deletion	F / F,P,P	The Traite's deletions were the oxygen pair's christening; the panel prices them there — E1's pre-fixed answer, the dissent expected and printed.	F stands; the expected dissent printed at full weight
Caloric • embedding	P / P,F,F	Carnot's theorem and Fourier's equation survive verbatim; caloric recovered no predecessor of its own — the two faces of one composite cell.	P stands; both faces printed; the two-faces fork recorded
Caloric • founders	F / F,P,P	Carnot's private notes surrendered conservation by 1832; published 1878, after his death — never tendered, never costly.	F stands; the split external verdict printed
Thermo • constant	P / P,P,P,F	J the unit bridge, demoted to a definition; k minted by the consolidation — one family reads the dual, the other the strict failure.	P stands — bracketed from both sides; the middle band confirmed as a middle
Thermo • derivation	S / S,S,P	Mayer 1842 from gas data in print; Helmholtz 1847; Joule 1843-49 — the simultaneous-discovery record is the over-determination.	S stands; the two-roads stipulation carries; the split printed
Thermo • formalism	S / S,P,S	Clapeyron's cycle re-read by Clausius without conservation; Boltzmann's probabilistic meaning on Clausius's entropy.	S stands; the depth fork recorded
Thermo • founders	P / P,S,S	Kelvin formed in caloric physics, publicly surrendered 1849-51, rebuilt his own scale's foundation — the coin tendered and paid; the depth the contestable half.	P stands under E2's pre-fixed graduation; the S-reading printed at full weight

Table 7 — The adjudication ledger: seventeen cells tried, three revised to duals, one confirmed, thirteen defended with their forks recorded — every revision by the pre-registered motion rule, no revision against it.

6. The Verdict: The Census After Eight Readings

Table 8 prints the census as it stands after eight readings. The dead side carries five theories and five duals' worth of printed faces: phlogiston's unification and miasma's deletion join the record this round; the ether's embedding keeps its direction-split composite; the slope row remains the compound every reading has ratified. The instrument's amendment map — the census's own error bars — now lists four domain-general discriminators and one domain-marked, five duals, and six named forks: ownership on the constant, reader-ownership on formalism, the birth-window on deletion, law-versus-vocabulary on unification, the graduation on resistance, and the two-faces split on embedding. Every fork is measured — each carries its reading record, and each record now includes at least two external administrations. The census has stopped pretending its instrument is seamless; it prints the seams and their widths.

Signature point	Phlogiston	The ether	Epicyles	Miasma	Caloric
Deletion at birth	F — Becher's principle renamed; nothing deleted	P * — the wave program deleted the corpuscle; its own center an inherited medium reinstated	F — extended Eudoxus; circularity never touched	F within the birth-window / P for the tradition at large — the Hippocratic deletion, dual (revised this run)	F — the fire-principle renamed and quantified; the Traite's deletions were the oxygen pair's christening
Unification	S as originally worded — the grouping survived into oxygen chemistry / P under the law-v s-vocabulary amendment — a posited substance is not a dissolved joint, dual (revised this run)	S — one medium under optics and electromagnetism	P * — one toolkit across the planets; vocabulary, not law	P — the zymotic class: administrative, not natural	P — one substance's bookkeeping; the substance-versus-motion division begged
Universal constant	F — no quantitative core; weight forced phlogiston negative	F — c belonged to the fields and outlived the medium (the ownership fork: three external dissents, recorded)	F — dozens of parameters, zero invariants	F — nothing constant; fitted coefficients per epidemic	F — material properties; Dulong-Petit the exception-ridden regularity the successor explained
Derivation-first	F — each new gas retrofitted, after the fact	P * — real derivational firsts; the defining note dishonored in 1887	P — parameters from centuries of records: content, never structure	F — the incumbent; data broke it rather than being derived	P — Carnot's theorem survived the frame; the friction anomalies retrofitted

Signature point	Phlogiston	The ether	Epicycles	Miasma	Caloric
Conservative embedding	F — deleted without remainder by oxygen chemistry	P * — the equations survived (Fresnel's coefficient as the first-order limit); the medium recovered as no limit — direction: F	P — devices survive as Fourier; the ontology deleted	F — the surviving sanitation was craft; no theorems existed to survive	P — the machinery survives verbatim (Fourier, Carnot); no predecessor recovered (the two-faces fork, recorded)
Formalism / meaning	F — no formalism; no measurable ever minted (the operationalized face; the aliens' as-written partial recorded)	S as written * — discriminates only operationalized (A6); the reader-ownership fork recorded	S as written * — the operationalized form discriminates; the reader-ownership fork recorded	F — no symbolic surface; no unit of miasma ever named	P — minted (Black's latent heat, the caloric, the calorimeter); the killing coin was the successor's
Founders' resistance	F — Priestley defended, paid nothing; defense is not payment	P * — Lorentz surrendered detectability, kept the medium	P * — Ptolemy's equant: a founder's compromise, collected for fourteen centuries	F — refusal at the price of the body itself	F — Carnot's private recantation, published 1878, was never tendered
Generative slope	S then D — the gas family; then negative weight	S then D — Fresnel to the interferometer; then the incoherence triangle	S then D — fourteen centuries; then the equant wound	S then D — the sanitary acts and the register; then the restatements	S then D — calorimetry to Carnot; then the capacity epicycles against Joule
Crisis-chaining	P — opened the air questions that killed it	S as written * — closed the corpuscular crisis, opened the drift problem; demoted regardless	P — the equant objection opened its execution	F — it faded; the archive was apparatus, not doctrine	P — the motive-power question, opened inside the program, became the successor's foundation

Table 8 — The census matrix after eight readings: five dead theories, five duals, six named forks. Revisions this run in full text; fork annotations carry their reading records in Table 6.

The living side, Table 9, changes by one annotation: germ theory's founders' cell becomes the record's fifth dual, and thermodynamics' middle band gains its bracket. Nothing else moves, and the stillness is the

finding: after three external readings — two families, three administrations, ninety cells each — the living side's satisfied cells stand where Volume I wrote them, with Newton's formalism dual the single, best-attested exception, and the two second-residents' partials (germ theory's constant, thermodynamics' constant and founders) exactly where the domain-marked and graduated amendments put them before any alien read a word.

Signature point	Germ theory (living)	Thermodynamics (living)
Deletion at birth	S — the double deletion: spontaneous generation and foul-air etiology	S — the double deletion: the substance of heat and unlimited convertibility
Unification	S — the brewer's vat and the surgical ward, one cause-class	S — heat and work one currency; the firebox and the falling weight the same phenomenon
Universal constant	F — domain-marked: no constant to install; the point abstains outside physics	P — the J/k dual, now bracketed by two external families: one reads the dual, the other the strict failure — the middle band confirmed as a middle
Derivation-first	P * — promissory notes honored; the founding carried by its own new experiments	S — over-determined by three roads (Mayer's derivation, Helmholtz's synthesis, Joule's measurements)
Conservative embedding	P — the practice recovered, the ontology deleted	S — Carnot verbatim, Fourier untouched, the no-work sector recovered with the reason why (the aliens' ontology-standard dissent recorded as the fork)
Formalism / meaning	F as written / S operationalized — the organism minted: stainable, countable, cultivable; the external readings fill the span's middle at P	S — absolute temperature, the joule, entropy minted; the efficiency bound a discriminating list that returns negatives
Founders' resistance	F as originally worded — the founders defended and won / P under the graduation — Pasteur's tendered Liebig coin, 1857-73, dual (revised this run)	P — Kelvin's tendered coin: formed in the caloric physics, publicly surrendered 1849-51, his own scale's foundation rebuilt (the depth fork recorded)
Generative slope	S — virology, immunology, antibiotics, molecular genetics: still generating	S — Gibbs, the third law, the quantum, Shannon's entropy, black holes: the deepest cross-domain harvest on record
Crisis-chaining	S — closed the etiological deadlock; opened the host, the filterables, the carrier state	S — closed the motive-power question; opened time's arrow, 175 years and counting

Table 9 — The living residents after eight readings: one new dual on germ theory's founders' row; thermodynamics' middle band bracketed; every other cell where it stood before any alien read a word.

What eight readings did to the genetic-dependence limit is now on the record in full. Amendment A4b confessed it; A4c ordered the alien reading and the infrastructure refused; A4d accepted the first delivery

and measured the offset at 3.3 points; this amendment receives two more deliveries — one from the returning family, one from a family never met — and finds them identical to the first on every shared cell. The limit dies as a threat not because the census won an argument but because the argument stopped being available: the verdicts are over-determined by readers this environment cannot select, cannot see, and cannot instruct. What remains of lineage is a calibration offset of three and a half points, monotone, zero inversions in three hundred and eighty-four cross-family comparisons, mapped cell for cell onto the instrument's own declared elastic cells. The census's error bars are its amendments, printed; its replication record is its defense, filed.

The standing offer remains standing, and it travels with this record. The kit is unchanged at ten theories — the verbatim prompt, the panel, the scale, the evidence rule, the forbidden list — and any third family, any human historian, any reader of any lineage can execute it; the convergence record printed here is the prior they will be measured against, and the brackets on the fifth pair are the cells their reading will fill. The census does not ask to be believed. It asks to be run — again, by anyone, under rules locked in advance — and prints what returns. The ledger is open.

72 shared-panel cells on which both external families return the first alien's verdict exactly seventy-two cells, zero exceptions, twice — the instrument's verdicts over-determined by readers this environment cannot select	4 cells where the two families part company all inside the fifth pair, all one step, all pre-registered as contestable before either family ran	0 distant disagreements in eight readings no external family has ever placed an S where the family places an F, or an F where the family places an S — the lineage effect is monotone
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The two readings' own numbers: the offer accepted twice more, the outgroup converging above the family band, and the census's elastic map confirmed by a second family that had never seen the instrument before.